

ACTIVITY 13 — FACILITATOR GUIDE**Cost Estimating, Budget and Reserves**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO2	3 - Plan the Project

Purpose of this activity

By the end of this activity learners will be able to select the right estimating technique for the information you actually have, and state its accuracy range.

What learners produce

cost estimates, cost baseline, reserve analysis — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Build the bottom-up estimate from the WBS. Create artifacts/13-cost-estimates.md. Take effort from the Lab 11 WBS dictionary and apply the differentiated resource rates.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
2	Contingency reserve versus management reserve. This is the central concept of the lab and one of the most reliably tested distinctions in the entire cost area. Learn it as a table and be able to reproduce it.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
3	Size the contingency reserve by expected monetary value. A reserve set as "10% because that is what we always do" is not an estimate; it is a habit. Size it from the risk register.	Ask what the number would have to be before the decision flips. That is the exam's favourite question.

4	Build the time-phased cost baseline. A total budget is not a baseline. A baseline is time-phased - it says how much should have been spent by any given date, which is what makes variance measurable. Plotted cumulatively it produces the characteristic S-curve: slow at the start while the team ramps and designs, steep through the build, flattening at the end as work completes.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
5	Funding limit reconciliation and cash flow. Contoso's board releases funds quarterly:	Circulate and ask each group to justify one entry against the case study facts.
6	Analyse the cost of quality	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Answer the exam-style scenarios	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.

Suggested timing

Minutes	Phase
9 min	Brief the scenario, confirm the inputs and assign roles
33 min	Work the first half of the steps individually, then pool as a group
34 min	Complete the remaining steps and finalise the deliverable
14 min	Group share-back, facilitator debrief and exam-style linkage

Total: 90 minutes

Discussion & decision prompts

1. Select the right estimating technique for the information you actually have, and state its accuracy range — how did your group decide, and what would change that decision?
2. Build a bottom-up estimate from the Lab 11 work packages that reconciles to the charter — how did your group decide, and what would change that decision?
3. Distinguish contingency reserve from management reserve precisely enough to answer any exam question on either — how did your group decide, and what would change that decision?
4. Your bottom-up labour total is SGD 347,980 and reconciles to the charter's SGD 348,000 within SGD 20 — can every member of your group defend it?
5. Where would cost estimating, budget and reserves have saved you time or money on a real project you have worked on?

What good looks like

- Your bottom-up labour total is SGD 347,980 and reconciles to the charter's SGD 348,000 within SGD 20.
- No role is funded from two budget lines. Check the UX designer and content lead in particular.
- You can state, without hesitating, that the cost baseline equals work plus contingency and the project budget equals cost baseline plus management reserve - and which reserve the PM controls.
- Your contingency reserve is derived from residual EMV, not from a percentage, and you can show why using gross EMV would double-count.
- Your residual net EMV is approximately SGD 26,600 and you stated the SGD 600 variance against the charter rather than rounding it away.
- Your cumulative PV at week 14 is exactly SGD 268,000 and your periods sum to exactly SGD 480,000. Lab 20 depends on both figures.

- Your S-curve is shallow, then steep, then flattening. A straight line means you divided rather than phased.
- You chose one contingency convention and applied it consistently, and can explain the phantom variance that mixing them produces.